

ACHINTYA RAI

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EDUCATION

University of California, San Diego
Bachelor of Science in Computer Engineering

Sept. 2024 – June 2028
GPA: 3.9/4.0

WORK EXPERIENCE

Adobe
Software Engineering Intern

San Jose, CA
June 2026 – Sept. 2026

- Diagnosed a **26-hour** processing outage via hourly volume analysis across **1,272** monitoring segments, setting the primary validation benchmark for a proactive anomaly detector.
- Engineered a time-series anomaly detector for multi-layer seasonality across **30+** countries.
- Automated an MLOps pipeline retraining and running inference across **10+** forecasting workflows.

Decklar (formerly Roambee)
Software Engineering Intern

Sunnyvale, CA
June 2024 – Aug. 2024

- Developed real-time shipment monitoring modules for **Amazon, Coca-Cola, and Samsung**, optimizing Python data pipelines to boost throughput **15%** across **1M+** weekly events.
- Implemented data structures for analytics modules, cutting reporting latency by **20%**.
- Enhanced operational workflow efficiency **15%** through Panels UI refinements and user testing.

VALT Health
Founding Software Engineer

Remote
Aug. 2025 – Feb. 2026

- Engineered a Flask backend ingesting **100k+** datapoints from wearable sensor devices, designing database schemas to unify formats across **20+** distinct sources.
- Implemented async batch processing in GCP, cutting redundant writes **60%** and raising speed **35%**.
- Automated environment setup and data seeding via Bash scripts, cutting testing time up to **80%**.

RESEARCH

Graph Theory & Neural Architecture
Undergraduate Research Fellow (NSF REU)

San Diego, CA
March 2026 – Present

- Engineered a Python pipeline to analyze model embedding manifolds, building extraction utilities that shrank **500GB** checkpoints to **13GB** localized datasets.
- Optimized preprocessing scripts for a **30x** runtime cut, enabling analysis on **8GB** RAM systems.
- Applied Louvain clustering to map model backbones, lifting network density to **13.25%**.

PROJECTS

Watchdog | *Python, Gemini API, Browser-Use, Asyncio, SQLite* 

April 2026

- Engineered an agentic browser monitor using Gemini Vision to intervene in real time.
- Built a concurrent Asyncio CLI for real-time URL blacklisting and task synchronization.

SquaredUpChess | *React, TypeScript, Django, WebSockets, AWS, Docker* 

Aug. 2025 – Dec. 2025

- Established a CI/CD pipeline with automated testing for **99.9%** deployment reliability.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL (Postgres/SQLite), Java, JavaScript, TypeScript, Bash, HTML/CSS

Development & Automation: Flask, FastAPI, Django, Node.js, Asyncio, WebSockets, REST APIs, PyTest, Git

Data, AI & Cloud: PyTorch, NumPy, TensorFlow, AWS, GCP, Docker, GitHub Actions, CI/CD, Linux